Resprucing update on tracking efficiencies and A/S week presentation

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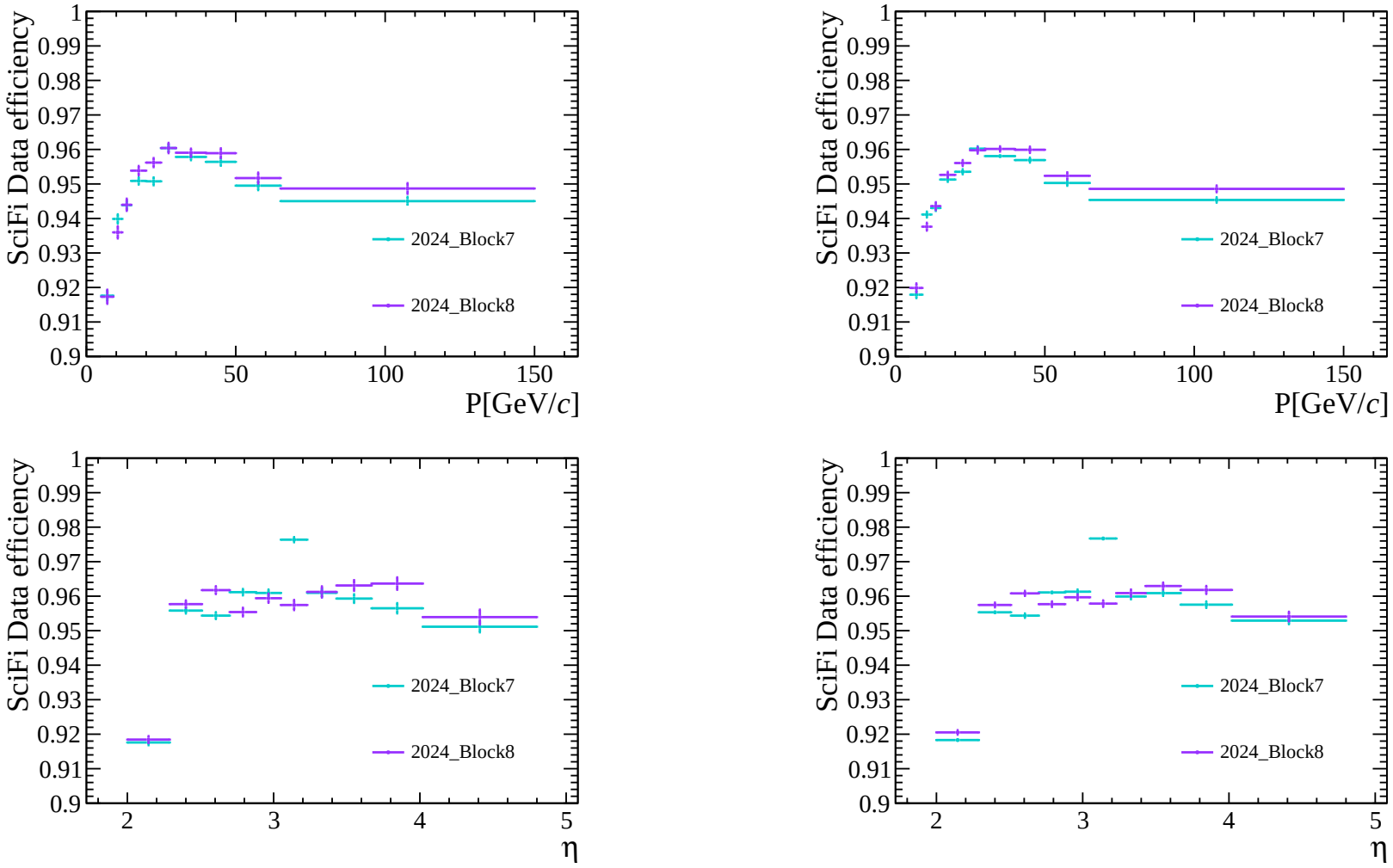
Resprucing of 2024 blocks 7 and 8

A resprucing campaign for 2024 blocks 7 and 8 was performed

- No sprucing prescale of 0.4
- I have run an AP and added the new samples to TrackCalib2
- Overall no big surprises
- Extra stats is nice, especially for MuonUT, but no significant changes

Before and after resprucing tracking efficiencies

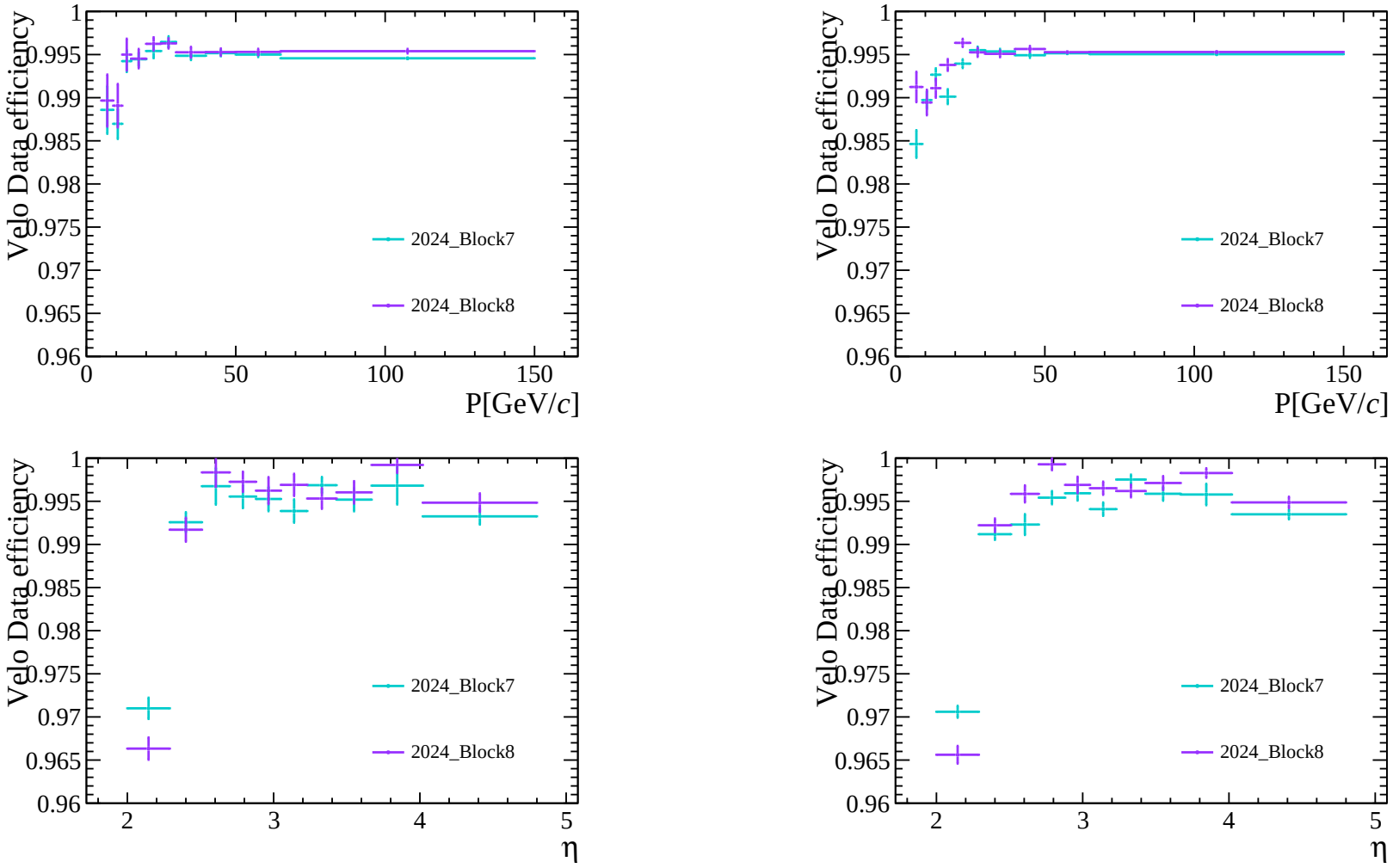
Comparison of tracking efficiencies in all data blocks



VeloMuon track efficiencies

Before and after resprucing tracking efficiencies

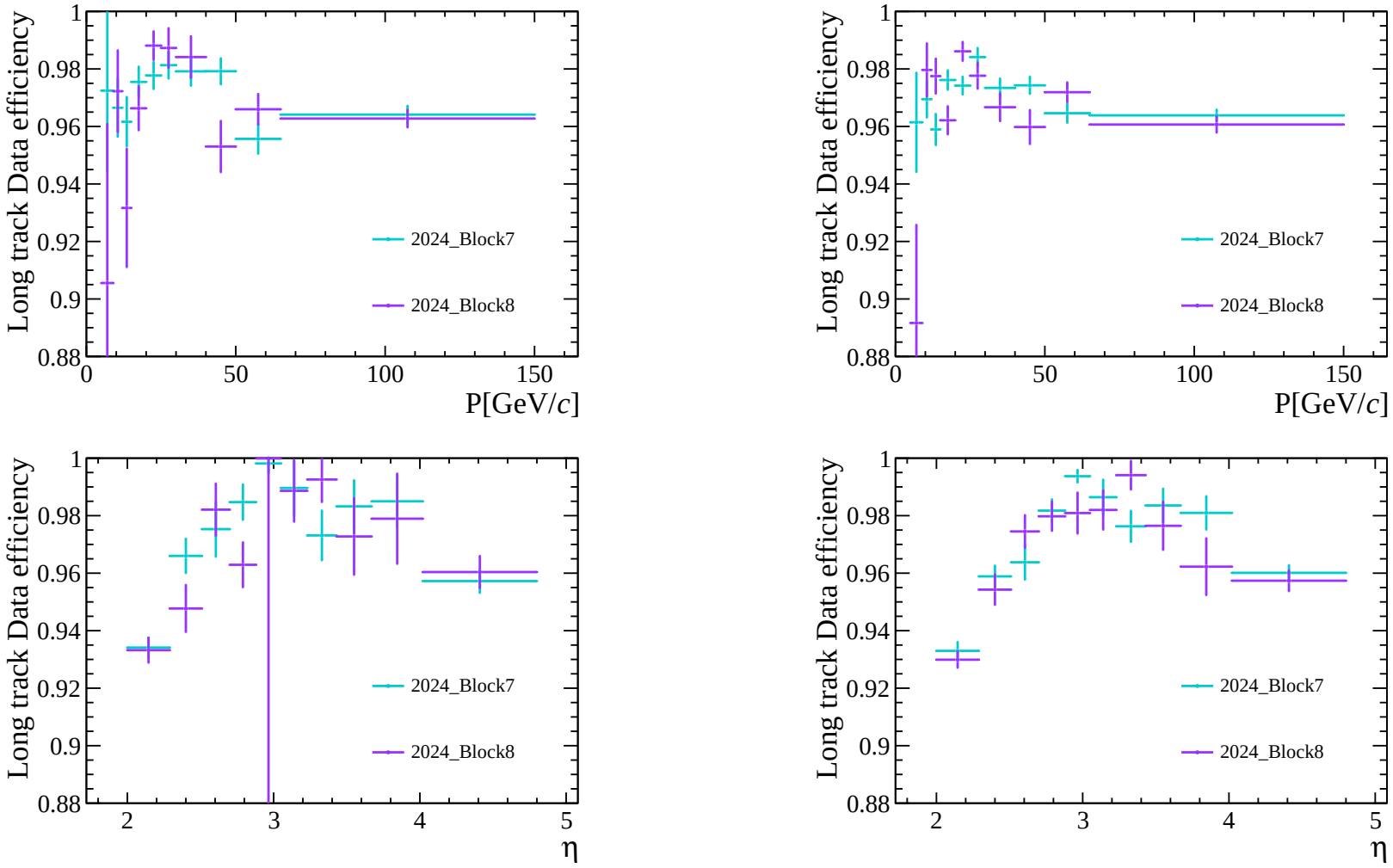
Comparison of tracking efficiencies in all data blocks



Downstream track efficiencies

Before and after resprucing tracking efficiencies

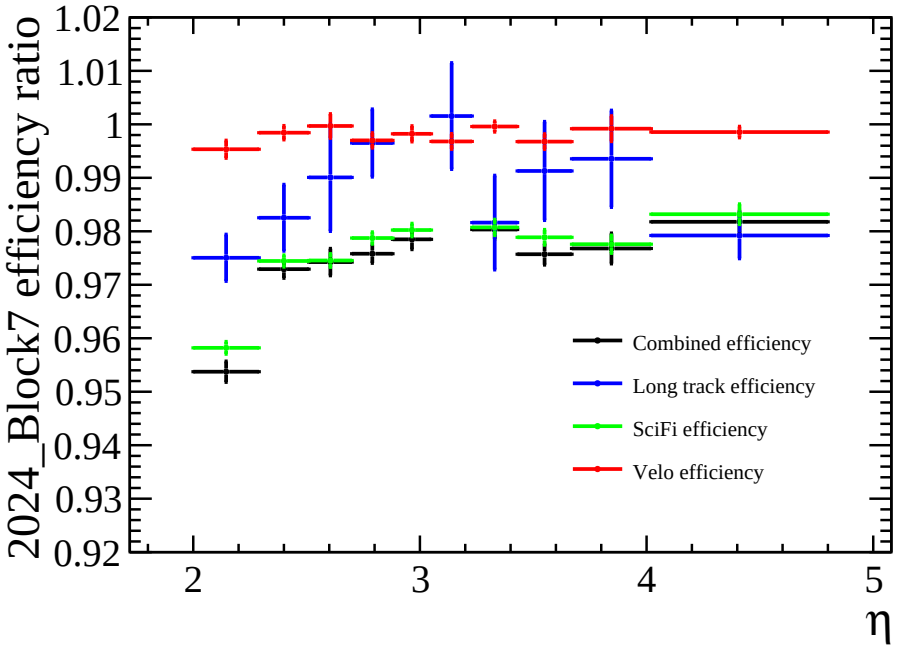
Comparison of tracking efficiencies in all data blocks

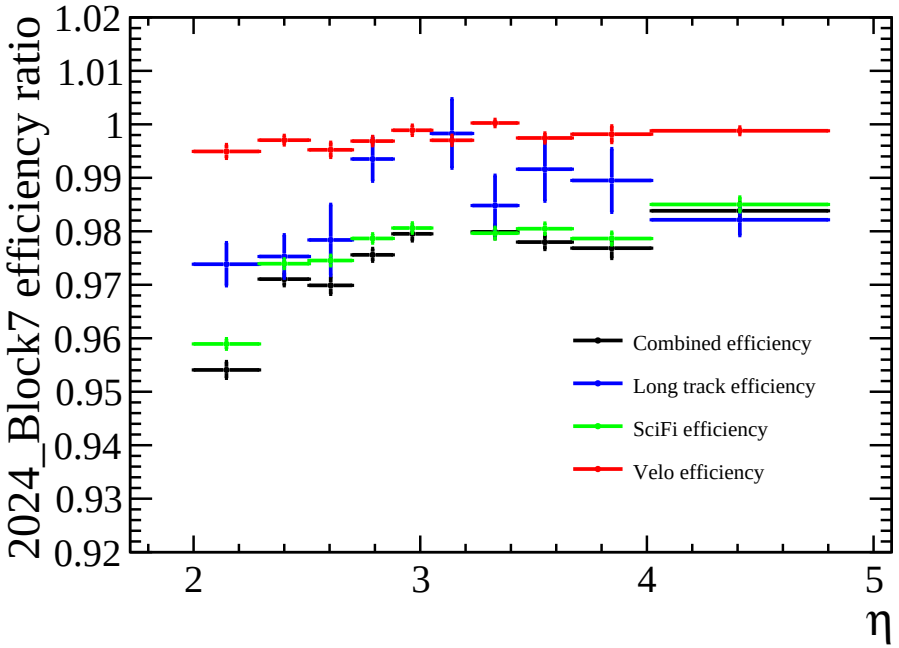


MuonUT track efficiencies

Before and after resprucing data/MC ratio

Data/MC ratio

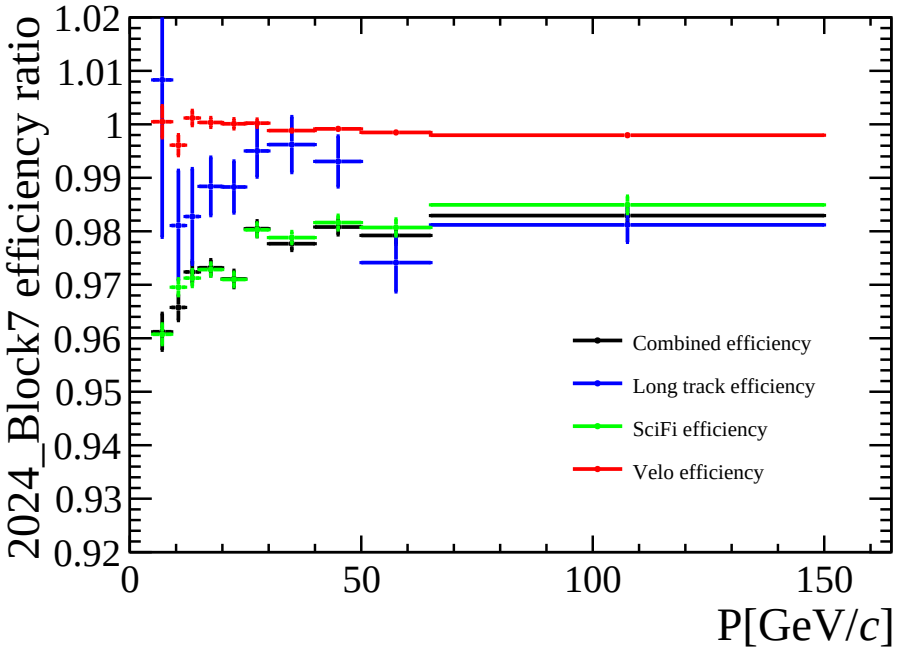


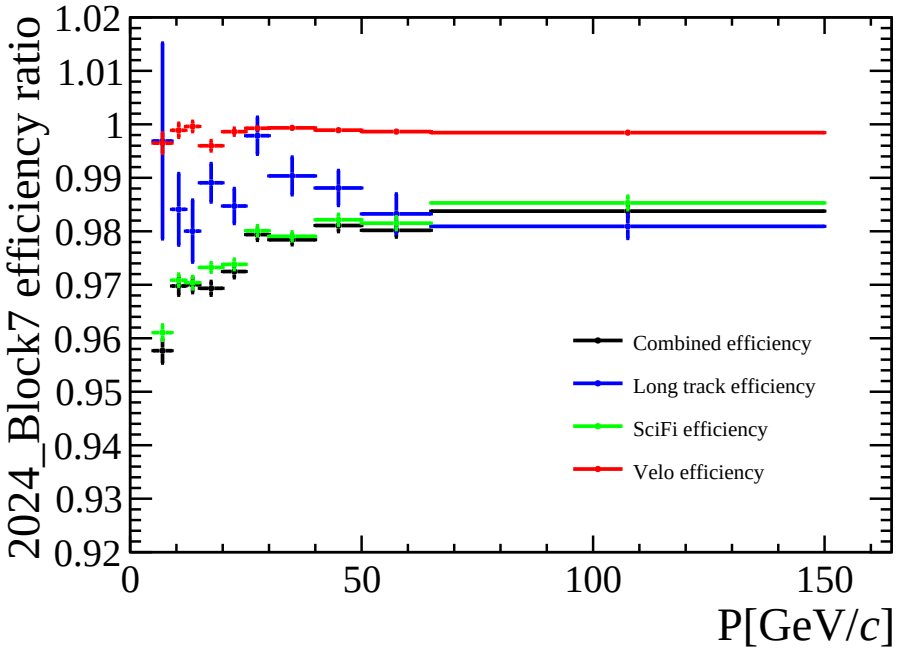


Data/MC ratio for Combined/MuonUT track efficiencies block 7 in η

Before and after resprucing data/MC ratio

Data/MC ratio

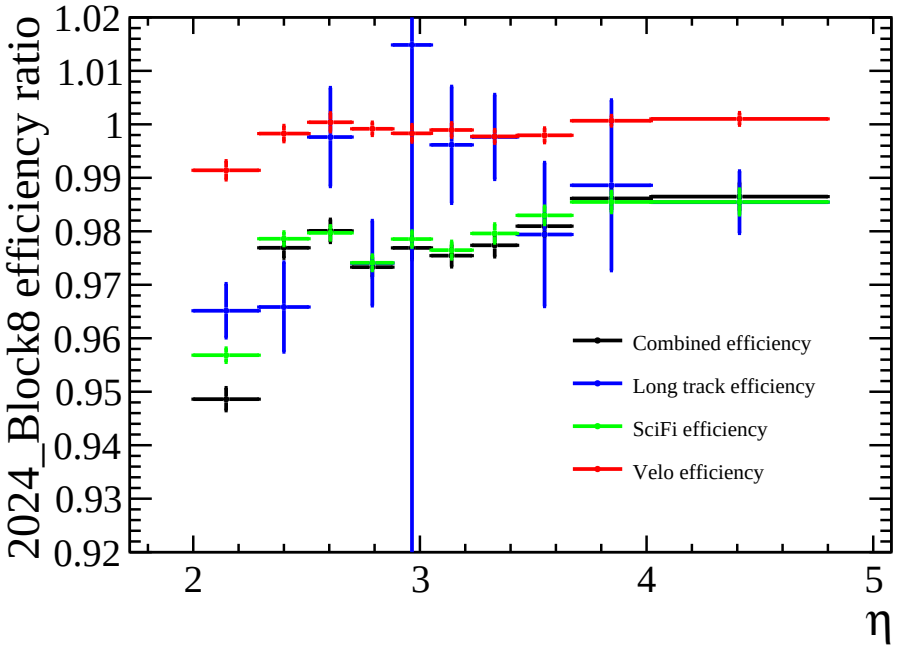


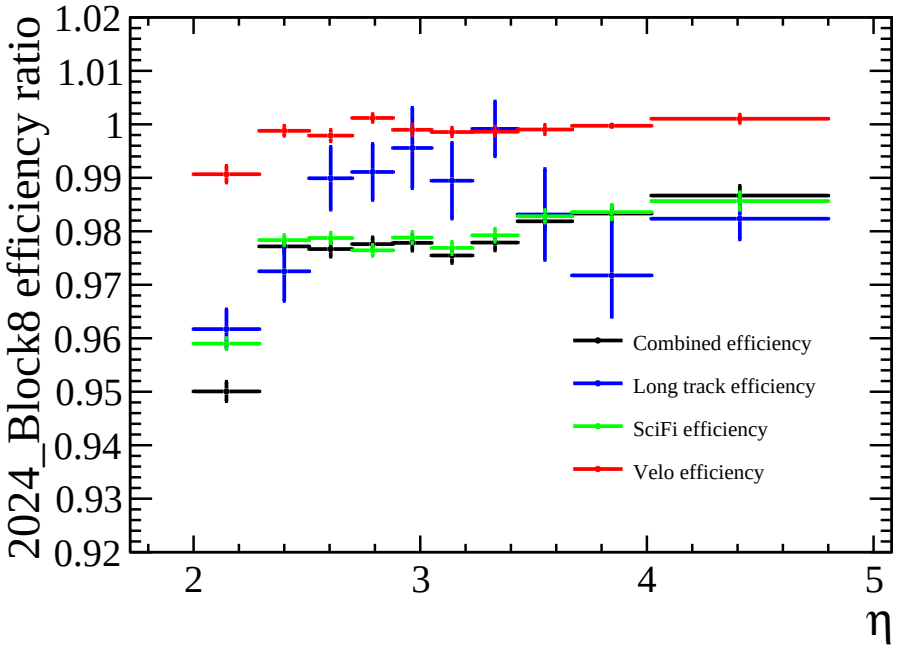


Data/MC ratio for Combined/MuonUT track efficiencies block 7 in p

Before and after resprucing data/MC ratio

Data/MC ratio

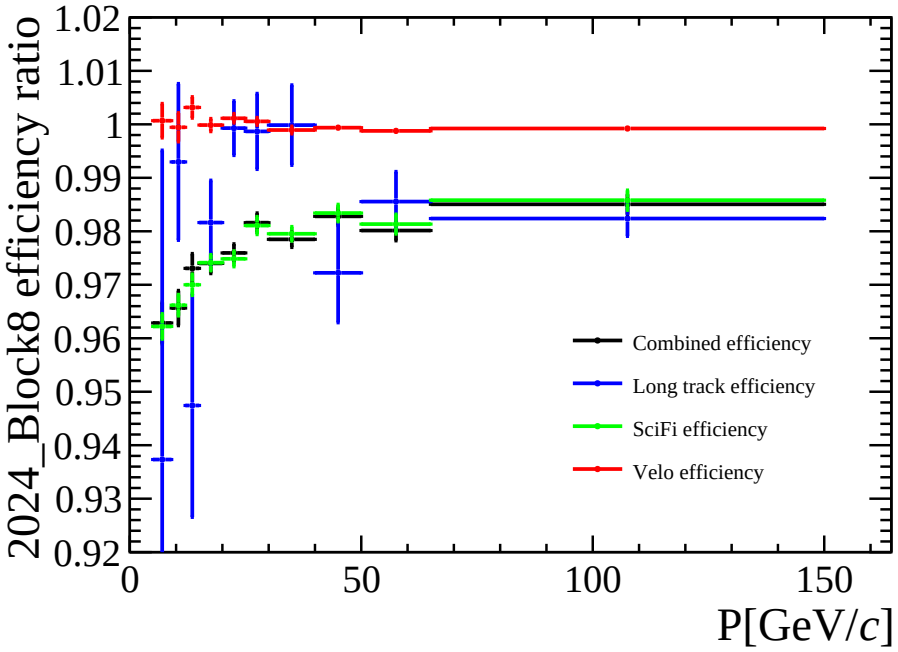


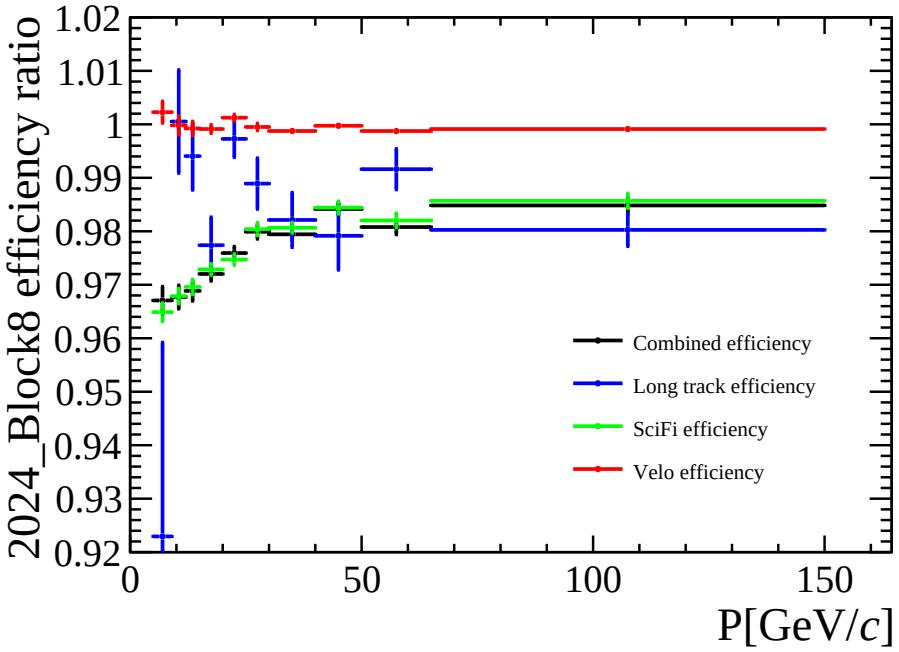


Data/MC ratio for Combined/MuonUT track efficiencies block 8 in η

Before and after resprucing data/MC ratio

Data/MC ratio





Data/MC ratio for Combined/MuonUT track efficiencies block 8 in p

A/S week presentation outline

Is the following outline for A/S week fine?

1. Introduction to track-and-probe method for tracking efficiencies
   - VeloMuon, Downstream, MuonUT
2. List the datasets and MC samples
3. Example of mass fit in TrackCalib2 to extract efficiency
4. Show 1D projections in p and η
5. Explain what we provide to analysts:
   - Data/MC ratio of tracking efficiencies
   - 2D binning in p and η
   - Statistical uncertainties from data/MC
   - Systematic uncertainties, mainly from Combined-MuonUT discrepancy
6. Introduce 2D binning and GitLab release of corrections
7. Discuss status of 2025 and block 1b?
8. Present method for hadronic corrections
9. Show hadronic correction results
10. Anything else to discuss on hadronic corrections?